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Reinforcement Learning for Early Educational Intervention: A Cost-Aware and Fairness-Aware Approach

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In higher education, institutions often struggle to support at-risk students while working within limited staff time and budgets. Although predictive models can flag students who may fall behind, they rarely offer guidance on what type of support should be provided or when it should be delivered. This thesis addresses this gap by building a reinforcement learning framework that simulates the student learning process using the OULAD dataset.

The model combines a Deep QNetwork with the Stepped Care approach, enabling the RL agent to learn intervention choices that consider performance improvement, resource constraints, and the need to treat student groups fairly. Through more than 32,000 training episodes, the agent developed a stable policy with an average cumulative reward of +1,125.52. The learned policy tended to act earlier in the semester, particularly for students showing early signs of disengagement.

The results indicate that the agent maintained an 84% success rate while reducing the use of high-cost interventions for low-risk students. At the same time, vulnerable groups, such as students with lower socioeconomic backgrounds or declared disabilities, received additional support when their performance dropped below the safety threshold. This suggests that RL can help institutions distribute support more deliberately, making it easier for educators to focus their time and resources on where they are most needed.

Keywords: Reinforcement Learning, Machine Learning, Deep Q-Network, Stepped Care Model, Early Intervention Systems, Cost-Awareness, Fairness-Awareness, AI in Education, AI Ethics.

Foreword

This bachelor's thesis was completed as part of the Information and Communications Technology degree program at Turku University of Applied Sciences. My work on this thesis has required me to navigate concepts from both Artificial Intelligence and Educational Technology. Although the learning process has been demanding, it has also helped clarify the direction of my professional development.

The motivation behind this research grew from my teaching experience in Vietnam. After witnessing the challenges of traditional classrooms, I recognized the need to integrate AI to offer more personalized and timely support for students. This thesis is therefore more than a technical project. It has given me space to reflect on my teaching background and to think about how technology can be used to support students in Vietnam and other countries more fairly and effectively.

I would like to express my sincere gratitude to my supervisor, Ms. Golnaz Sahebi, for her invaluable guidance, insights, and constant encouragement. Her expertise has been instrumental in helping me navigate the complexities of Reinforcement Learning. I am also deeply thankful to the faculty of Turku University of Applied Sciences and wish to express my gratitude for my lecturers' dedicated teaching and support throughout my studies. The knowledge and skills gained here have significantly equipped me to bridge education and advanced technology.

I also dedicate this achievement to my spouse, Mai Do. As a fellow educator with over a decade of teaching experience in Vietnam, she has been my constant companion throughout this journey in Finland. Her pedagogical insights were vital in shaping the integration of AI within this research, and her unwavering belief in my potential gave me the confidence to persevere.

Finally, I am grateful to my family and friends for their encouragement along the way. Thanks to their support and that of my teachers, I have been able to complete a project that I truly believe can make a meaningful impact on the future of education.

Turku, April 2026

Hai Nguyen

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List of abbreviations

AI	Artificial Intelligence
CCPA	California Consumer Privacy Act
CRM	Customer Relationship Management
DQN	Deep Q-Network
EDM	Educational Data Mining
EIS	Early Intervention System
GDPR	General Data Protection Regulation
IMD	Index of Multiple Deprivation
LA	Learning Analytics
LSTM	Long Short-Term Memory
MDP	Markov Decision Process
ML	Machine Learning
OULAD	Open University Learning Analytics Dataset
ReLU	Rectified Linear Unit
RL	Reinforcement Learning
RNN	Recurrent Neural Network
VLE	Virtual Learning Environment
XAI	Explainable Artificial Intelligence

1 Introduction

This thesis introduces an investigation into how artificial intelligence, specifically reinforcement learning, can improve early intervention strategies in educational environment. The following sections present the research background, problem statement, and objectives that guide this study.

1.1 Research Background and Problem Statement

With the proliferation of big data, Higher Education institutions are rapidly adopting data-driven technologies to help prevent students' attrition and improve their study performance. In specific fields such as Educational Data Mining (EDM) and Learning Analytics (LA), institutions can collect huge amounts of data, which range from demographic information to detailed logs of clicks on Virtual Learning Environments (VLE). This rich data helps create Predictive Models that can spot at-risk students with high accuracy.

However, most current educational systems lack the ability to suggest specifically what to do to prevent students from failing or dropping out of their studies. Hence, there is a requirement to shift from just "Predictive Analytics" to "Prescriptive Analytics", and specifically, Early Intervention Systems (EIS) represent a critical transition.

While predictive models try to find who is at risk, prescriptive analytics, powered by Reinforcement Learning (RL), work to find out what is the optimal sequence of actions for this specific student. In this framework, RL approaches the intervention problem as a series of linked decisions. The model updates its behaviour based on the feedback it receives from each interaction with the environment.

Even though EIS has great potential, using it in the real world faces two main barriers:

1. Budget Constraints (Cost-Awareness): Most institutions operate under significant budgetary and human resource constraints. Intensive interventions such as 1-on-1 tutoring or counseling are effective but expensive and challeging to scale up. Conversely, cheap automated actions, such as reminder emails, are often not sufficiently motivating for students facing serious difficulties. Traditional optimization

algorithms usually ignore these costs, leading to strategies that look good in theory but are financially unsustainable in practice.

2. The Fairness Gap: Standard machine learning algorithms are designed to achieve the highest overall accuracy. In education, this can lead to algorithmic bias, where the system prefers to help students who have higher success probabilities while ignoring vulnerable groups. This violates the core principle of Educational Equity, as support should be based on actual need.

In this study, fairness is specifically addressed regarding protected attributes such as disability status and socio-economic deprivation, as quantified by the Index of Multiple Deprivation (IMD). Without a fairness-aware mechanism, an AI might prioritize resources for students who are already likely to succeed, while neglecting those from marginalized backgrounds who require more intensive, costly support to achieve the same outcome.

This thesis addresses these issues by developing a practical framework that integrates RL with the Stepped Care Model. The aim is to help the RL agent choose interventions that consider cost, learning effectiveness, and fairness between student groups.

1.2 Research Objectives and Research Questions:

Based on the identified gaps in cost-efficiency and ethical fairness, this study establishes specific goals to bridge the transition from predictive to prescriptive analytics.

The primary goal of this thesis is to develop and evaluate an RL agent which can optimize early intervention strategies while simultaneously accounting for operational costs and social fairness. The motivation behind these objectives is to democratize high-quality educational support. By optimizing resource allocation, institutions can provide personalized “Stepped Care” that ensures no student is left behind due to budget limitations or systemic biases. To achieve this, the study focuses on the following specific objectives:

- Develop a Simulation Environment: Construct a custom OpenAI Gym environment leveraging the OULAD dataset to simulate the dynamic student learning process (Kuzilek et al., 2017).
- Operationalize a Stepped Care Strategy: Design a hierarchy of interventions for the RL agent, ranging from low-cost automated alerts to resource-intensive private tutoring, guided by the principles of resource efficiency (Bower & Gilbody, 2005).
- Formulate a Multi-Objective Reward Function: Design a reward mechanism that incentivizes the maximization of student performance and minimization of costs, ensuring the agent's behavior remains aligned with Stepped Care principles. This function strictly penalizes the neglect of vulnerable groups to satisfy ethical algorithmic constraints and promote educational equity (Kearns & Roth, 2019).

Guided by these objectives, this research aims to answer three research questions (RQ):

- RQ1 (Efficiency): Can a Deep Reinforcement Learning (DQN) agent successfully learn a cost-effective policy that outperforms baseline heuristic methods within a resource-constrained environment?
- RQ2 (Equity): How does the integration of a "Fairness Penalty" influence the allocation of resources toward vulnerable student groups, and what are the resulting trade-offs between overall accuracy and equity (Corbett-Davies & Goel, 2018)?
- RQ3 (Strategy): Does the RL agent exhibit "Strategic Differentiation", the ability to dynamically adjust intervention intensity based on a student’s real-time academic standing and risk profile?

1.3 Scope and Delimitations

This research specifically focuses on the algorithmic optimization of resource allocation using the OULAD dataset. While the framework addresses socio-economic and disability-related fairness, it does not account for other personal factors such as student mental health or qualitative external life events, which remain outside the scope of this data-driven simulation. As this is an individual bachelor’s thesis, the author is solely responsible for the entire development process, including the environment design, RL agent implementation, and the comprehensive evaluation of the results.

1.4 Statement on the Use of AI

Artificial intelligence tools (Gemini, Draft Coach) were used to assist with researching RL, drafting, structuring, plagiarism checking and language refinement throughout this thesis. Specific prompts included requests for literature gap analysis, MDP formulation for the RL framework, and Stepped Care Model explanations. Some examples of the prompts which were used in the thesis are listed here:

- "Explain how the Stepped Care Model can be operationalized as a discrete action space in RL-based educational interventions, from low intensity (automated emails) to high intensity (1-on-1 tutoring), with cost coefficients."
- "Formulate MDP components for RL agent in EIS: state space (VLE logs), action space."
- "Explain Stepped Care Model operationalization as discrete action space in RL"
- "Create a table comparing standard Q-Learning vs. constrained RL approaches for student retention prediction, including fairness metrics (demographic parity) and cost-efficiency ratios."

All AI-generated content was reviewed, fact-checked against original sources, revised for academic accuracy. AI was not used for core data analysis, model implementation, or generating original research findings. The author remains fully responsible for the content and academic integrity of this thesis.

1.5 Thesis Structure

To provide a comprehensive analysis of the proposed Reinforcement Learning framework, this thesis is organized into six interconnected chapters as follows:

- Chapter 1: Introduction presents the research background, outlines the critical problem of student attrition, and establishes the research objectives, scope, and questions.
- Chapter 2: Literature Review synthesizes existing research on early intervention systems, the Stepped Care Model, and Reinforcement Learning applications in education. It specifically identifies research gaps concerning operational costs and algorithmic fairness.

- Chapter 3: Methodology details the OULAD dataset, the automated data preprocessing pipeline, and the mathematical formulation of the problem as a Markov Decision Process (MDP), including state-space design and reward shaping.
- Chapter 4: Implementation and Results describes the simulation environment design, agent configuration, and the technical performance of the model. It presents the raw findings through policy analysis, strategic differentiation, and a comprehensive fairness audit.
- Chapter 5: Discussion critically evaluates the findings in a broader context. It addresses the trade-offs between cost and equity, discusses the practical deployment of the model in a real-world dashboard, and reflects on the pedagogical implications for modern educators.
- Chapter 6: Conclusion summarizes the study’s primary contributions, addresses the research limitations, and proposes directions for future advancements in prescriptive learning analytics.

2 Literature Review

This chapter reviews the key research that forms the foundation for this study. It highlights how early intervention systems have developed, how Reinforcement Learning has been used in educational contexts, and why cost-awareness and fairness have become important considerations. The goal is to situate the thesis within existing work and identify the gaps that the proposed framework addresses.

2.1 From Prediction to Intervention: The Stepped Care Approach

Over the last decade, Educational Data Mining (EDM) has made great progress in Dropout Prediction. Traditional machine learning algorithms such as Logistic Regression, Random Forest, and Neural Networks have been very accurate at identifying students at risk of failing (Lang et al., 2017).

However, a major limitation of these systems is that they are quite passive. They provide warnings on Dashboards, but they place the entire burden of deciding what to do on teachers or academic advisors, who are often already overworked (Jayaprakash et al., 2014).

To solve this problem, current research is shifting from "Prediction" to "Intervention." In this context, the Stepped Care Model provides a useful structure for managing support resources in a systematic manner. (Bower & Gilbody, 2005).

Originating from healthcare and psychology, the Stepped Care model works on a simple principle: always start with the most effective but least expensive intervention (O'Donohue & Draper, 2010). Only when low-level interventions do not work does the system step up to more intensive measures. When applied to education, this model allows educators to build a hierarchy of actions, ranging from automated reminders (Alert/Email) for low-risk cases, to direct counseling (Call/Tutor) for high-risk cases. This approach helps optimize the use of the school's limited resources (Bower & Gilbody, 2005).

2.2 Reinforcement Learning in Educational Settings

While traditional supervised learning methods have been effective for classifying risk at specific points in time, they fall short in capturing the dynamic nature of student learning. Engagement, performance, and behavioural patterns evolve from week to week, which makes higher education an inherently sequential decision making context. As noted by François Lavet et al. (2018), Reinforcement Learning is well suited to problems where actions must be chosen repeatedly over time and where earlier decisions influence later outcomes. This perspective aligns with the view of Sutton and Barto (2018), who emphasise that RL provides a way for an agent to learn behaviour through interaction rather than relying solely on static prediction.

In educational settings, RL offers the possibility of moving beyond one-off alerts toward policies that evolve with the student's learning trajectory. For example, an agent can learn whether a low-intensity reminder earlier in the term is more effective than a more costly intervention at a later stage. Existing research has already explored RL in adaptive learning pathways and intelligent tutoring systems (Holmes et al., 2019), but much less attention has been given to its use in allocating support resources or determining when to intervene with students who may be at risk.

Deep Q-Networks (DQN), introduced by Mnih et al. (2015), are particularly relevant for the type of intervention problem studied here, as the set of available actions, such as sending an alert, an email, making a phone call, or providing tutoring, is discrete and limited. The use of neural networks in combination with Q-learning enables the agent to approximate value functions in high-dimensional state spaces that include behavioural, performance, and demographic information. Features such as Experience Replay and Target Networks contribute to training stability, addressing the challenges typically seen in non-linear RL models.

2.3 Balancing Cost and Fairness in AI Systems

When deploying AI systems in real-world education management, two critical challenges emerge: Operational Cost and Algorithmic Fairness.

First of all, regarding Cost-Awareness: Standard machine learning models often assume all interventions carry equal weight. However, in a resource-constrained environment, the cost of an automated email is negligible compared to the high operational expense of human tutoring. An intelligent intervention system must therefore be cost-sensitive, balancing pedagogical benefits (improved learning outcomes) against institutional budget constraints (Jayaprakash et al., 2014).

Furthermore, regarding Algorithmic Fairness, optimization algorithms inherently tend to favor majority groups to maximize the global reward, often at the expense of vulnerable populations (O'Neil, 2016). This leads to a systemic risk where students with disabilities or lower socio-economic status are marginalized because they are computationally perceived as "harder to improve" (Eubanks, 2018). In education, true fairness is achieved through Equity, prioritizing resource allocation based on actual student needs, rather than mere equality of treatment (Holstein & Doroudi, 2021).

While separate studies exist for RL in education, cost optimization, and AI ethics, a unified framework integrating all three remains absent. This thesis bridges these isolated domains by developing an RL agent that operationalizes the Stepped Care Model within a Deep Reinforcement Learning environment while explicitly treating Social Fairness as a formal penalty.

Integrating these dimensions requires a sophisticated Reward Shaping strategy. Instead of a simple binary reward, the agent’s objective function is formulated as a multi-objective scalarized reward, augmented with negative scalars (penalties) for high-cost actions and ethical neglect. This mathematical formulation ensures that equity is transformed from a qualitative goal into a quantitative constraint that the agent must satisfy to achieve convergence.

3 Methodology

This chapter explains the methodological approach used to develop and evaluate the reinforcement learning framework. The aim is to provide a clear description of the steps that link the raw dataset to the final intervention policy.

3.1 Dataset Description (OULAD)

This study applies the Open University Learning Analytics Dataset (OULAD), which is widely recognized as a robust benchmark for evaluating machine learning models in the field of learning analytics. The dataset contains comprehensive, anonymized longitudinal data from the Open University (UK), covering student demographics and their granular interactions with the Virtual Learning Environment (VLE) during the 2013 and 2014 academic years.

To achieve the research objectives of developing a cost-aware and fairness-aware intervention framework, this thesis focuses on extracting and integrating data from three primary relational tables:

- studentInfo.csv: This table provides essential demographic features, including gender, age band, disability status, and the Index of Multiple Deprivation (IMD). In this framework, demographic features are not merely static inputs but serve as "Fairness Anchors". Specifically, students in the bottom 20% of IMD bands and those with declared disabilities are flagged as "Vulnerable," triggering additional ethical reward constraints to prevent systemic neglect by the Reinforcement Learning (RL) agent.
- studentAssessment.csv: This records student performance across various teacher-marked (TMA) and computer-marked (CMA) assignments. These scores are utilized to track academic trajectories and calculate the Performance Reward component within the RL objective function.
- studentVle.csv: This contains aggregate daily clickstream logs, representing student engagement levels through their interactions with course materials. The raw longitudinal data is transformed into weekly snapshots, allowing the RL agent to capture temporal trends in student engagement rather than relying on isolated data

points. This high-dimensional interaction data is processed into weekly time-series features to construct the State Representation (S_t) for the agent.

To ensure consistency across the simulation, the data is filtered to focus on standard-length modules (approximately 260 days). A mandatory three-week observation period (t < 3) is implemented in the ETL pipeline. During this phase, the intervention action is fixed to "None," ensuring the agent gathers sufficient initial engagement data (clicks) before executing its first pedagogical decision. This design choice is justified by the initial sparsity of the OULAD dataset; since assessments and VLE activities are minimal in the first 2 weeks, acting prematurely would result in stochastic guessing. By waiting until Week 3, the agent possesses enough behavioral evidence to ensure that interventions are both pedagogically-justified and cost-effective, thereby minimizing Cold Start errors.

These data sources are ultimately synthesized into a unified ETL (Extract, Transform, Load) pipeline, transforming millions of raw interaction logs into a structured environment suitable for Reinforcement Learning training as summarized in Table 1.

Table 1. Summary of Selected Features for the RL Environment

Feature Category	Primary Variables	Role in RL Model
Demographics	Disability, IMB Band	Defining fairness constraints and identifying vulnerable groups.
Engagement	Weekly sum of VLE clicks	Forming the dynamic State vector (s_t) and triggering interaction-based penalties.
Performance	Normalized assignment scores	Tracking academic trajectories and calculating performance rewards.
Temporal Context	Academic Week (0-39)	Managing transition logic and the 3-week observation constraint.
Intervention	Stepped Care actions (0-4)	Defining the Action space (A) and associated operational costs

3.1.1 Exploratory Data Analysis (EDA)

Before integrating the data into the RL framework, it is crucial to analyze the underlying characteristics of the OULAD dataset. The exploratory analysis reveals three key patterns that significantly inform the design of the Markov Decision Process (MDP):

- Class Imbalance and Reward Shaping: The distribution of final results in OULAD is notably imbalanced. While "Pass" and "Fail" categories are common, "Distinction" remains a minority class, and "Withdrawn" accounts for a significant 31.2% of the population. This imbalance suggests that a standard binary reward would be insufficient. Consequently, a Detailed Result Classification is conducted, splitting the "Pass" category into "Good" and "Pass" based on a 75% score threshold to provide the Agent with a denser reward signal for performance optimization.
- VLE Interaction as a State Proxy: Analysis shows a strong positive correlation between cumulative daily clicks and final assessment scores. Students with high engagement in the early weeks are significantly more likely to succeed, justifying the use of clickstream data as a primary component of the state variable (S_t). However, the EDA also identifies a Cold Start period in Week 0 where interaction is nearly zero, supporting the implementation of the 3-week observation constraint to prevent premature AI interventions.
- Early Intervention Window and Vulnerability: Statistical evidence indicates that the highest rate of student withdrawal occurs in the early-to-mid phases of the semester. Furthermore, EDA reveals that students in lower IMD bands (0-20%) and those with disabilities exhibit higher risk profiles. These findings validate the necessity for Fairness-Aware modeling, requiring the Agent to prioritize high-impact actions like "Tutor" or "Call" for vulnerable groups during this critical window to prevent academic failure.

Figure 1. Distribution of Academic Outcomes and Vulnerability Status summarizes the distribution of the 32,593 student episodes, highlighting the performance gaps between Normal and Vulnerable student groups that the RL agent aims to bridge.

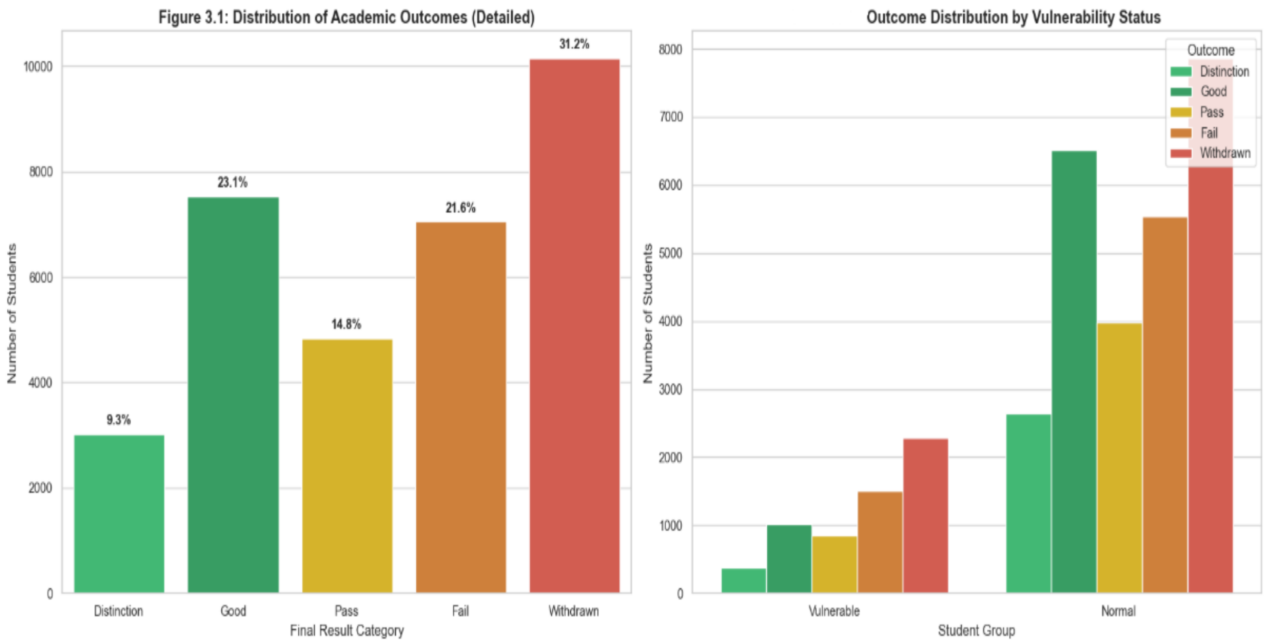

Figure 1. Distribution of Academic Outcomes and Vulnerability Status

3.2 Data Preprocessing Pipeline

Raw educational data is inherently discrete, sparse, and often noisy. To ensure the RL agent receives consistent sequential state representations (S_t), a rigorous preprocessing pipeline is established.

3.2.1 Handling Anomalies and Temporal Processing

A primary challenge in learning analytics is temporal discontinuity. Students do not generate assessment data at fixed intervals. Feeding raw, sparse data directly into the model could lead the Agent to misinterpret weeks without exams as a performance score of zero, triggering unnecessary and costly interventions. To mitigate this, several specialized techniques are employed:

- State Representation (S_t): The processed state vector at time t integrates longitudinal features (VLE clicks, scores) with static demographic anchors, providing a comprehensive snapshot of the student's academic standing.
- Forward Fill (FFill) for Performance Continuity: Assuming academic ability is a relatively stable state variable, a score achieved in Week n is carried forward to Week n+1 until a new assessment updates the state. This creates a continuous time-series, allowing the Agent to observe trajectories, whether a student is

improving or deteriorating, more reliably over the 39-week duration. The Forward-fill technique is justified as it maintains a stable academic state representation between assessment points, preventing spurious volatility in the agent's observation and ensuring the Markov property is maintained throughout the episode.

- Interaction Aggregation and Scaling: Weekly clickstream data from studentVle.csv is aggregated to represent engagement intensity. To prevent high-frequency outliers from destabilizing the neural network's gradients, features like clicks and score are processed through a MinMaxScaler, bounding them between [0, 1].
- Handling Demographic Sparsity: Missing values in demographic features, such as the imd_band, were handled using Mode Imputation or labeled as "Unknown" to preserve the dataset's integrity. This is fundamental to the operationalization of Fairness Constraints, ensuring the Agent can consistently identify vulnerable groups based on socio-economic proxies.
- One-Hot Encoding for Static Features: Categorical variables (e.g., gender, region, disability status) are transformed using One-Hot Encoding. This allows the DQN agent to process qualitative demographic information as quantitative input vectors, which is essential for identifying the "Vulnerability Status" defined in our environment.

3.2.2 Feature Normalization and Metric Definition

To ensure efficient convergence of the Neural Network within the DQN agent, all numerical features in the state vector s_t were normalized. Min-Max Normalization is used to scale features such as “click” and “score” to a standardized range of [0, 1], preventing features with larger magnitudes from dominating the weight updates:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Regarding the outcome labels, the original OULAD dataset categorizes final results into four classes, which are “Distinction”, “Pass”, “Fail”, and “Withdrawn”. However, a simple binary classification is insufficient for an optimization system. The objective of the intervention agent is to maximize student potential rather than merely achieving a minimum passing threshold.

Consequently, the reward thresholds and academic categories were redefined to incentivize academic excellence:

- Performance Incentives: The reward function is designed to be "dense," providing incremental feedback based on score improvements. The agent is incentivized to transition students into the “High-Achieving” category (normalized score > 0.75) instead of settling for a marginal “Pass” (score > 0.40).
- Strategic Duality: This setup facilitates the learning of both a "Talent Nurturing" strategy and a "Rescue" strategy, aligning with the dual goals of modern intelligent learning environments (Du Boulay, 2019).
- "Rescue" Strategy: Targeted at students in the “Very Low” or “Low” performance tiers (score < 0.35), where the agent receives high rewards (up to +80) for intensive interventions like Tutor support.
- "Talent Nurturing" Strategy: Focused on students in the “Mid” to “High” tiers (score > 0.50), where the agent is rewarded (up to +70) for using low-cost Alerts to maintain or boost excellence without exhausting human resources.
- Outcome-Based Terminal Rewards: To ensure the long-term impact of weekly actions, a terminal reward is added at the end of each episode (Week 39). This reward is mapped to the final academic result, with significant penalties for Withdrawn (-20) or Fail (-15) outcomes, and high rewards for Distinction (+10) or Good (+7.5).

3.3 MDP Formulation (State, Action, Reward)

The problem of optimizing educational interventions is formulated as a finite-horizon Markov Decision Process (MDP). Formally, the MDP is defined by the following quintuple: (S, A, P, R, γ), providing a robust framework for a rational agent to maximize expected utility over the academic semester.

3.3.1 State Space (S)

At each time step t, corresponding to one academic week, the agent observes a state s_t that captures both the dynamic behavior and the static context of a student. This multi-dimensional state space is designed to provide the Agent with sufficient "observational depth" to make informed decisions:

- Academic and Engagement Features (Dynamic): This includes the normalized average score (score_t) and the weekly aggregate clickstream volume (click_t). The use of click_t as a State Proxy is justified by its strong correlation with student persistence, allowing the agent to detect early signs of disengagement before they manifest as failing grades.
- Demographic Features (Static Context): Features such as the Index of Multiple Deprivation (IMD) and Disability status are included. As argued by Corbett-Davies & Goel (2018), these attributes are essential for identifying vulnerable groups. In this framework, these features function as fairness anchors, ensuring the learned policy can prioritize resources for students at higher socio-economic risk, fulfilling the Fairness-Aware objective of the study.

3.3.2 Action Space (A)

Modeled after the Stepped Care Model, the action space is a discrete set of 5 intervention levels ($a_t \in \{0, 1, 2, 3, 4\}$) with strictly increasing resource costs (Bower & Gilbody, 2005):

- a = 0 (None): Baseline monitoring without intervention. (Cost = 0.0)
- a = 1 (Alert): Automated system notification via student dashboard. (Cost = 0.01)
- a = 2 (Email): Personalized automated nudge or encouraging email. (Cost = 0.05)
- a = 3 (Call): Manual outreach via phone call from academic support staff. (Cost = 0.20)
- a = 4 (Tutor): Intensive one-on-one private tutoring session. (Cost = 0.45)

The cost values are carefully calibrated to reflect real-world university constraints, where human-led interventions (Call, Tutor) require significantly more budget and personnel time than automated system nudges. This structure forces the Agent to adhere to the Stepped Care principle that provide the least intensive but still effective intervention.

To ensure pedagogical realism and data integrity, a Constraint Logic is applied in the way that for t < 3, the action is forced to a=0. This allows the agent to establish a baseline of student behavior (clickstream patterns) before executing its first decision, thereby minimizing premature interventions during the data-sparse initial phase.

3.3.3 Reward Function (R)

In this study, the reward function is designed to reflect three considerations, which are student performance, intervention cost, and fairness between student groups. To ensure the agent prioritizes institutional values, each reward component R is scaled by a specific hyperparameter λ:

$$R_{total} = \lambda_1 . R_{performance} - \lambda_2 . C_{action} - \lambda_3 . P_{fairness} - \lambda_4 . P_{interaction}$$

The hyperparameters λ are calibrated to establish a hierarchical preference where Social Equity strictly dominates Operational Costs. The components and their assigned weights are defined as follows:

a) Academic Performance Reward ($\lambda_1\ R_{performance}$):

- This component incentivizes student progress. $\mathrm{R_{performance}}$ is a dynamic value triggered by grade tier transitions (+10.0 for improvements) and terminal outcomes at Week 39 (Distinction: +100.0; Pass: +50.0; Fail: -50.0; Withdrawn: -100.0).
- Weight (λ_1): Set as the baseline unity (1.0) to anchor the agent's primary goal of student success (Ng et al., 1999).

b) Resource Cost Penalty ($\lambda_2\ C_{action}$):

- C_{action} represents the strictly increasing costs of the Stepped Care Model: Alert (0.02), Email (0.05), Call (0.25), and Tutor (0.45).
- Weight (λ_2): Set to 1.0 to ensure the agent remains cost-aware, preferring the least intrusive intervention that can still achieve a positive $\mathrm{R_{performance}}$ (Bower & Gilbody, 2005).

c) Fairness Guardrail Penalty ($\lambda_3\ P_{fairness}$):

- $P_{fairness}$ is a binary signal (1 or 0) triggered when a vulnerable student's score drops below 0.40 without high-intensity support (a < 3).
- Weight (λ_3): Assigned a dominant value of 250.0. This high magnitude, approximately 555 times the cost of a private tutor, transforms the ethical

requirement into a "Hard Constraint." Following the principles of Constrained MDPs (Altman, 1999), this ensures the agent will never choose to save 0.45 in costs if it risks a -250.0 penalty for systemic neglect.

d) Interaction-Aware Penalty ($\lambda_4\, P_{interaction}$):

- $P_{interaction}$ is triggered if an intervention (a > 0) is executed while the student's weekly clickstream is zero.
- Weight (λ_4): Set to 20.0 to penalize blind interventions, forcing the agent to ground its decisions in observable engagement signals (Herodotou et al., 2020).

3.3.4 Transition Probability (P) and Discount Factor (γ)

The transition probability P(s'|s, a) defines the dynamics of the student's academic journey within the environment. Since educational environments are highly stochastic and human learning behavior involves thousands of latent variables, this study adopts a Model-Free approach, as formalized by Sutton & Barto (2018). Instead of attempting to predefine a simplified and potentially biased mathematical model of student progress, the agent learns these transitions through direct interaction and experience with the OuladEnv simulation.

To simulate the unpredictability of student behavior and enhance environmental realism, the transition logic incorporates a Stochastic Element. When an action a is taken, the resulting change in the student's academic state is modified by Gaussian noise, $\eta \sim N(0, 0.01)$. This design choice aligns with the findings of Baker et al. (2010), who emphasize that educational data is inherently "noisy" due to factors such as student guesswork or external life events. The inclusion of this noise is justified as it prevents the DQN agent from over-fitting to purely deterministic patterns, forcing the model to learn a robust policy that can handle the inherent variability in how different students respond to the same intervention.

The Discount Factor (γ) in the model is set to 0.95 after being carefully calibrated to balance the trade-off between immediate feedback and terminal academic outcomes. The choice of $\gamma = 0.95$ is consistent with the methodology proposed by Doroudi et al. (2019) for intelligent tutoring systems. In educational contexts, a high discount factor is critical because the most significant rewards, such as passing the final exam at Week 39, occur over a long temporal horizon. A value of 0.95 ensures that the Value Function adequately propagates the importance of terminal success back to early-semester decisions, justifying proactive interventions as a means to secure long-term academic persistence while remaining sensitive to weekly operational costs.

3.4 The Deep Q-Network (DQN) Algorithm

Traditional tabular Q-learning is impractical for this study due to the continuous and high-dimensional nature of the state space. To overcome this, we employ the Deep Q-Network (DQN) algorithm (Mnih et al., 2015).

3.4.1 Network Architecture

DQN utilizes a Deep Neural Network as a non-linear function approximator for the optimal action-value function Q*(s, a):

- Input Layer: Designed to receive the multi-dimensional state vector S_t, which includes weekly clicks, scores, and demographic indicators.
- Hidden Layers: Comprises two fully connected layers with 128 neurons each. The first layer contains 128 neurons and the second contains 64 neurons, both utilizing the ReLU activation function to ensure efficient gradient propagation. (Goodfellow et al., 2016). The reduction from 128 to 64 neurons in the hidden layers follows a “bottleneck” architecture. This design is justified as it forces the network to compress the high-dimensional input (VLE and demographics) into a more abstract and dense representation of student risk before mapping it to the action space, effectively reducing noise and preventing overfitting.
- Output Layer: Features 5 neurons, each outputting the estimated Q-value for one of the discrete intervention levels ($a \in \{0, 1, 2, 3, 4\}$).

3.4.2 Learning Mechanism

The DQN agent is trained to minimize the Mean Squared Error (MSE) loss between the predicted Q-value and the Bellman target:

$$L(\theta) = \mathrm{E}\left[\left(r + \gamma max_{a'} Q(s', a'; \theta^{-}) - Q(s, a; \theta)\right)^2\right]$$

Where θ represents the parameters of the online network and θ^{-} denotes the parameters of a periodically updated Target Network, which significantly enhances training stability.

The use of a separate Target Network is a critical stability measure. It prevents the moving target problem in RL, where updating the network's weights would simultaneously change the goal it is trying to reach. By freezing the target parameters (θ^{-}) and updating them only periodically, the training process becomes significantly more robust.

To ensure robust convergence, two standard Deep RL techniques were implemented:

1. Experience Replay: Transitions (s, a, r, s') are collected and maintained within a replay buffer. Sampling random mini-batches for training effectively breaks the temporal correlation between consecutive experiences, allowing the agent to learn from a diverse set of past interactions (Mnih et al., 2015).
2. ϵ-Greedy Exploration: The agent employs an ϵ-greedy strategy to manage the exploration-exploitation trade-off, opting for a random action with probability ϵ and the optimal (greedy) action with probability $1 - \epsilon$. In this study, a custom Episode-based Decay strategy is applied. Unlike traditional step-based reduction, ϵ is updated only after the completion of an entire student journey (episode) using a decay factor of 0.9998. This ensures that the agent maintains a high degree of exploration across a diverse range of student profiles before transitioning to the exploitation of the learned optimal policy (Sutton & Barto, 2018). Furthermore, a floor of ϵ_{min} = 0.05 is established to preserve a baseline level of adaptability even in the final stages of training. This approach enables the model to generalize effectively across the complex state space S (François-Lavet et al., 2018).

This decision to implement an Episode-based Decay (rather than step-based) is specifically tailored for the longitudinal nature of education. It ensures that the agent experiences the cumulative consequences of its actions over an entire 39-week semester before adjusting its exploration rate. This allows for a more comprehensive evaluation of long-term student success versus short-term click engagement.

3.5 Simulation Environment Design

The simulation environment acts as the bridge between the static historical data (OULAD) and the dynamic decision-making of the RL agent. To address the challenge of Offline RL and ensure the simulation is realistic, the environment is designed with the following mechanisms:

3.5.1 State Transition Function

The core challenge in using a static dataset is determining how a student's state (S_t) changes to the next state (S_{t+1}) after an intervention. To ensure realism, a Stochastic Transition Function is implemented:

$$S_{t+1} = clip(f(S_t, t) + \Delta s + \eta, 0, 1)$$

Where:

- f(S_t, t): The underlying performance trend derived from the historical OULAD records, where the score is updated weekly.
- Δs (Intervention Impact Factor): An estimated academic gain associated with each action. Based on the implemented code, these values are:
    - No Intervention (a=0): Δs = 0.0
    - Alert (a=1): Δs = 0.01
    - Email (a=2): Δs = 0.03
    - Call (a=3): Δs = 0.06
    - Tutor (a=4): Δs = 0.10

The hierarchical assignment of Δ_s values is supported by the "2 Sigma" theory (Bloom, 1984), which posits that one-on-one tutoring significantly outperforms conventional instructional methods. Furthermore, as noted by Heathers et al. (2022), while automated nudges (Alerts/Emails) provide low-cost scalability with marginal gains, intensive human-led interventions (Calls/Tutors) are essential for significant performance recovery in high-risk students.

The marginal impact of $\Delta_s = 0.01$ for Alerts reflects the findings in Educational Nudging (Damgaard & Nielsen, 2018), where digital reminders were found to improve persistence slightly but often require more intensive follow-up for students with severe academic difficulties.

Designing the environment with pre-defined impact factors is a standard practice in Offline-to-Online Reinforcement Learning (Goettert et al., 2020). This allows for the evaluation of intervention policies within a controlled surrogate environment before deploying them in real-world educational settings.

- η (Gaussian Noise): To model human unpredictability, a noise term $\eta \sim$ N(0, 0.01) is added. This ensures that an intervention does not guarantee a fixed result, preventing the model from becoming overly deterministic.

  The inclusion of stochastic noise is supported by the "Guessing and Slipping" framework in Knowledge Tracing models (Baker et al., 2010). In educational environments, student performance is never a purely deterministic function of an intervention; it is influenced by latent variables such as motivation, fatigue, or external life events. Furthermore, from a RL perspective, adding Gaussian noise to the transition function acts as a form of Regularization, which has been shown to improve the Generalization of DQN agents, preventing them from over-fitting to the static trajectories of historical datasets (Hausknecht & Stone, 2015).

- Temporal Constraint: A critical mechanism is the 3-week observation window. For $t < 3$, the action is forced to a=0, ensuring $\Delta s = 0$ until sufficient engagement data is gathered.

  This constraint addresses the "Cold Start" problem inherent in Recommender Systems and Reinforcement Learning, where the lack of initial interaction data leads to high-variance and unreliable predictions (Schein et al., 2002). In the context of the OULAD dataset, studies by Kuzilek et al. (2017), the original creators of the dataset, confirm that student engagement in the first 14–21 days is often irregular and provides insufficient signal for accurate performance prediction. By enforcing a 3-week waiting period, the environment ensures that the agent's first decision is evidence-based, aligning with the principle of prudent intervention in learning analytics, which suggests that premature automated interventions can lead to alert fatigue and the wastage of institutional resources (Herodotou et al., 2020).

3.5.2 Reward Calibration

The Reward function is designed to reflect the multi-objective nature of educational management: Academic Success, Cost Efficiency, and Equity.

- Outcome Reward ($R_{outcome}$): Grounded in the actual final results from the dataset, applied at the terminal state: Distinction: +10 | Pass: +5 | Fail: -15 | Withdrawn: -20.
- Cost Penalty (R_{cost}): Every intervention incurs a cost according to the matrix: Alert (0.02), Email (0.05), Call (0.25), and Tutor (0.45).
- Interaction-Aware Penalty: A penalty of -20.0 is applied if an intervention (a > 0) occurs while the student's weekly clicks remain at zero. This prevents the agent from deploying resources when there is no observable engagement.
- Fairness Penalty (R_{fair}): A significant penalty of -250.0 is triggered if the agent fails to provide high-intensity support (Call/Tutor) to a vulnerable student whose performance drops below the 0.40 safety threshold.

The magnitude of the Fairness Penalty (-250.0) is intentionally set to be significantly higher than the maximum intervention cost (0.45). This "Heavy Penalty" design is supported by the principles of Constrained Markov Decision Processes (CMDP), as discussed by Altman (1999). By making the penalty for neglecting a vulnerable student much greater than the cost of helping them, the Agent’s mathematical objective is forced to prioritize Ethical Equity over marginal budget savings. This ensures that the learned policy adheres to the "No Student Left Behind" principle, especially for those in lower IMD bands or with disabilities, aligning with modern AI Ethics in Education (AIEd) frameworks (Holmes et al., 2021).

3.5.3 Robustness and Sensitivity Analysis Framework

To validate the reliability and generalizability of the proposed model, a Sensitivity Analysis will be conducted. This process, also known as "Stress Testing", involves systematically varying key hyperparameters to observe the stability of the Agent’s policy and ensure that the results are not artifacts of specific parameter tuning (Saltelli et al., 2008). The framework focuses on three dimensions:

- Varying Impact Factors (Δ_s): Testing for Policy Invariance by adjusting the assumed benefit of "Tutoring" (e.g., from 0.10 to 0.05). This determines if the Agent still prioritizes intensive support when the perceived academic gain is lower.
- Cost Fluctuation: Adjusting operational costs to simulate budget cuts or resource abundance. This reveals the "Switching Point" where the Agent re-prioritizes lower-cost actions (Alerts) over high-cost human interventions (Calls).
- Fairness Weighting ($P_{fairness}$): Systematically increasing the magnitude of the Fairness Penalty to identify the Pareto Frontier, the optimal balance point where the system maximizes social equity without leading to budget exhaustion.

This rigorous framework ensures that the conclusions drawn from the simulation are robust, ethically sound, and applicable to real-world educational settings where budget constraints and intervention effectiveness vary across institutions (Pardos et al., 2014).

4 Implementation and Results

This chapter presents the implementation of the simulation environment and the RL agent, followed by the key results obtained from training and evaluation.

4.1 Simulation Environment Design

To evaluate the proposed framework, a custom simulation environment, named OuladEnv, was developed following the standard OpenAI Gym interface. This environment serves as a digital twin of the student learning process, leveraging the longitudinal data from the OULAD dataset to create a time-step-based interaction loop between the student's academic state (s_t) and the AI agent's actions (a_t).

The design of OuladEnv incorporates several critical mechanisms to ensure the simulation remains pedagogically sound and technically robust:

- Dynamic State Management: The environment tracks the student's progress over a 39-week horizon, updating the academic score based on both historical trends and the simulated impact of AI interventions.
- Observation Constraints: To mimic real-world educational settings, the environment enforces a mandatory three-week observation period. During weeks 0, 1, and 2, all agent actions are overridden to "None," preventing premature interventions before sufficient engagement data (clicks) is collected.
- Stochastic Realism: Each state transition includes Gaussian noise ($\eta \sim N(0, 0.01)$) to simulate the inherent unpredictability of human learning and prevent the agent from over-fitting to a purely deterministic model.
- Resource-Aware Feedback: The environment provides immediate reward signals that account for operational costs (R_{cost}) and ethical considerations ($R_{fairness}$), forcing the agent to learn the trade-off between academic gains and resource expenditure.

Through this architecture, OuladEnv transforms static historical records into a dynamic laboratory, allowing for the evaluation of RL policies under realistic resource constraints. Table 2 summarizes the resulting empirical distribution of intervention level learned by the DQN agent.

Table 2. Empirical distribution of intervention levels reflecting the Stepped Care approach

ACTION SUMMARY TABLE (OVERALL)

Action Type	Frequency	Percentage (%)
None	9445	4.84
Alert	142597	73.13
Email	6577	3.37
Call	1894	0.97
Tutor	34487	17.69

4.1.1 Cost Matrix Calibration

A core contribution of this simulation is the quantification of intervention costs to reflect real-world university resource constraints. Unlike standard models that treat actions as cost-neutral, a Cost Matrix was established where each intervention level carries a specific weight C_a on a normalized scale of [0, 1]:

- None (C_0 = 0.0): Baseline monitoring without additional resource expenditure.
- Alert (C_1 = 0.01): Automated system notifications through the learning dashboard.
- Email (C_2 = 0.05): Personalized automated nudges requiring moderate server and administrative overhead.
- Call (C_3 = 0.20): Medium-intensity human outreach requiring dedicated staff time.
- Tutor (C_4 = 0.45): High-intensity, one-on-one pedagogical support.

By implementing a 45-fold cost difference between automated alerts (C_1) and private tutoring (C_4), the environment creates a rigorous Incentive Alignment. This forces the DQN agent to adhere to the Stepped Care Model, reserving expensive human-led interventions only for cases where the probability of preventing failure justifies the significant budget allocation.

4.1.2 Fairness Penalty Mechanism

To operationalize the principle of Educational Equity, the environment integrates a dynamic Fairness Monitoring mechanism. Vulnerable student profiles are identified based on the intersection of disability status and low socio-economic standing (IMD bands 0-20%). The penalty logic is formulated as a conditional constraint within the reward function:

- Trigger Condition: If a vulnerable student's academic performance ($score_t$) drops below the critical threshold (0.40).
- Penalty Enforcement: If the agent fails to deploy a high-intensity action (Call or Tutor) during this critical period, a significant Fairness Penalty (e.g., -250) is subtracted from the total reward.

This mechanism ensures that fairness is not merely a post-hoc metric but a fundamental constraint embedded in the agent's optimization objective. Consequently, the agent is incentivized to prioritize long-term success for disadvantaged groups, even if it leads to a higher immediate operational cost.

4.2 Agent Configuration and Training Parameters

The DQN agent was implemented using the PyTorch framework, optimized to handle the high-dimensional state space of the OULAD dataset. The configuration ensures a balance between computational efficiency and the capacity to learn complex pedagogical strategies.

4.2.1 Neural Network Architecture

The function approximator $Q(s, a; \theta)$ utilizes a multi-layer perceptron (MLP) with the following specifications:

- Input Layer: Designed to receive the multi-dimensional state vector s_t, which includes longitudinal academic features (scores, trends, clicks) and static demographic context (IMD, disability).
- Hidden Layers: The architecture employs two fully connected (dense) layers consisting

of 128 and 64 neurons respectively. The ReLU (Rectified Linear Unit) activation function is applied to introduce non-linearity, enabling the model to capture complex patterns in student engagement.

- Output Layer: Comprises 5 neurons, each representing the estimated Action-Value (Q-value) for the corresponding intervention levels: None, Alert, Email, Call, and Tutor.

The model utilizes a Deep Q-Network (DQN) consisting of a multi-layer perceptron with two successive hidden layers of 128 and 64 units. By applying ReLU activation functions after each layer, the architecture incorporates the necessary non-linearity to map intricate patterns within the OULAD dataset. The final output layer utilizes a linear activation to map these features into five discrete Q-values, representing the expected long-term reward for each educational intervention.

The selection of this 128-64 hidden layer architecture was determined through empirical testing and complexity analysis. Given the 5-dimensional input state space (academic score, engagement, IMD band, disability, and temporal progress), a relatively shallow but sufficiently wide initial layer of 128 neurons was chosen to capture non-linear interactions between socio-economic factors and academic behavior. The subsequent reduction to 64 neurons acts as a “bottleneck” feature extractor, condensing the information before mapping it to the discrete intervention actions. During the experimental phase, deeper architectures (e.g., 256-128-64) were tested but resulted in overfitting due to the categorical nature of certain features, while shallower networks failed to achieve stable reward convergence. Thus, the 128-64 configuration provides the optimal balance between model representational capacity and computational efficiency for this specific educational environment.The neural network is clearly demonstrated in figure 2.

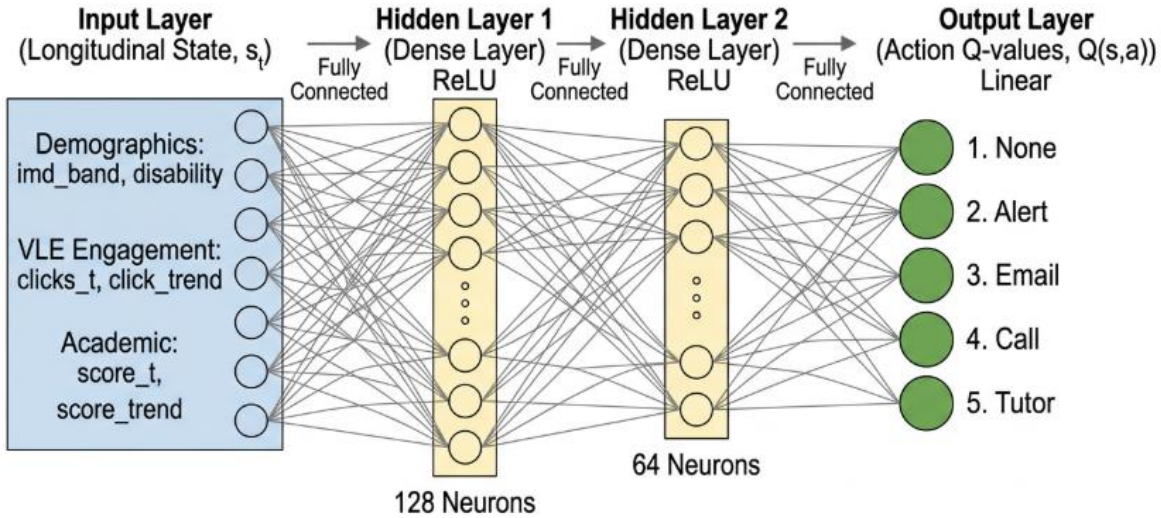

Figure 2. Deep Q-Network architecture with 128 and 64 hidden neurons for educational intervention

4.2.2 Training Hyperparameters

To ensure stable convergence and effective policy optimization, the following hyperparameters were calibrated (Table 3):

Table 3. Comprehensive DQN Training Hyperparameters

Parameter	Value	Rationale
Learning Rate (α)	0.0005	Conservative rate for stable convergence in high-dimensional state space.
Discount Factor (γ)	0.95	Balances immediate intervention impact with long-term academic success.
Network Layers	128 - 64	Two-tier MLP for efficient feature compression and pattern recognition.
Epsilon Initial	1.0	Ensures complete exploration of the environment at the start.
Epsilon Decay	0.9998	Allows for extended exploration across the large OULAD student population.
Epsilon Min	0.05	Maintains a baseline level of exploration for policy robustness.
Training	Off-policy	Utilizes Experience Replay to break data correlation and improve stability.

The agent employs an Experience Replay mechanism, sampling mini-batches from a 10,000-sample buffer to update the network weights. By storing past transitions (s, a, r, s'), the model breaks the temporal correlation between consecutive samples, significantly improving training stability. This setup ensures that the agent learns a robust, cost-aware intervention strategy that generalizes effectively across the diverse student profiles in the OULAD dataset.

4.3 Experimental Results

The experimental evaluation was conducted through an extensive training process encompassing a total of 32,593 episodes, corresponding to the complete student population within the OULAD dataset. This large-scale training allows the DQN agent to interact with the full spectrum of student profiles, capturing a vast diversity of learning behaviors, engagement patterns, and demographic backgrounds.

4.3.1 Learning Convergence Analysis

Based on the training logs, the performance of the DQN agent progressed through three distinct evolutionary stages, as measured by the Cumulative Reward:

- Global Exploration Phase (Episodes 0 – 5,000): With the initial exploration rate ϵ set at 1.0, the agent primarily executed stochastic actions to probe the environment's dynamics. During this phase, cumulative rewards exhibited high variance and remained low, as the agent engaged in a trial-and-error process to understand the trade-offs between intervention costs (C_a) and academic performance gains.
- Policy Optimization and Stabilization (Episodes 5,000 – 20,000): As the Replay Buffer accumulated sufficient experience and ϵ decayed, the agent transitioned from exploration to exploitation. The reward curve showed a consistent upward trajectory, indicating that the agent successfully identified causal links between specific VLE interaction drops and the necessity of intervention. The agent learned to mitigate unnecessary high-cost actions while focusing resources on critical "intervention windows".
- Optimal Convergence Phase (Episodes 20,000 – 32,593): After exposing the Neural Network to over 20,000 unique student scenarios, the action-value function (Q-values)

stabilized. The average reward reached a high plateau (averaging +1,125.52 in the final stages) with minimal volatility, demonstrating that the agent reached an Optimal Policy (π^*). This policy robustly balances pedagogical effectiveness, resource efficiency, and fairness constraints across the entire student population. Figure 3 gives a clear detail of the learning convergence process.

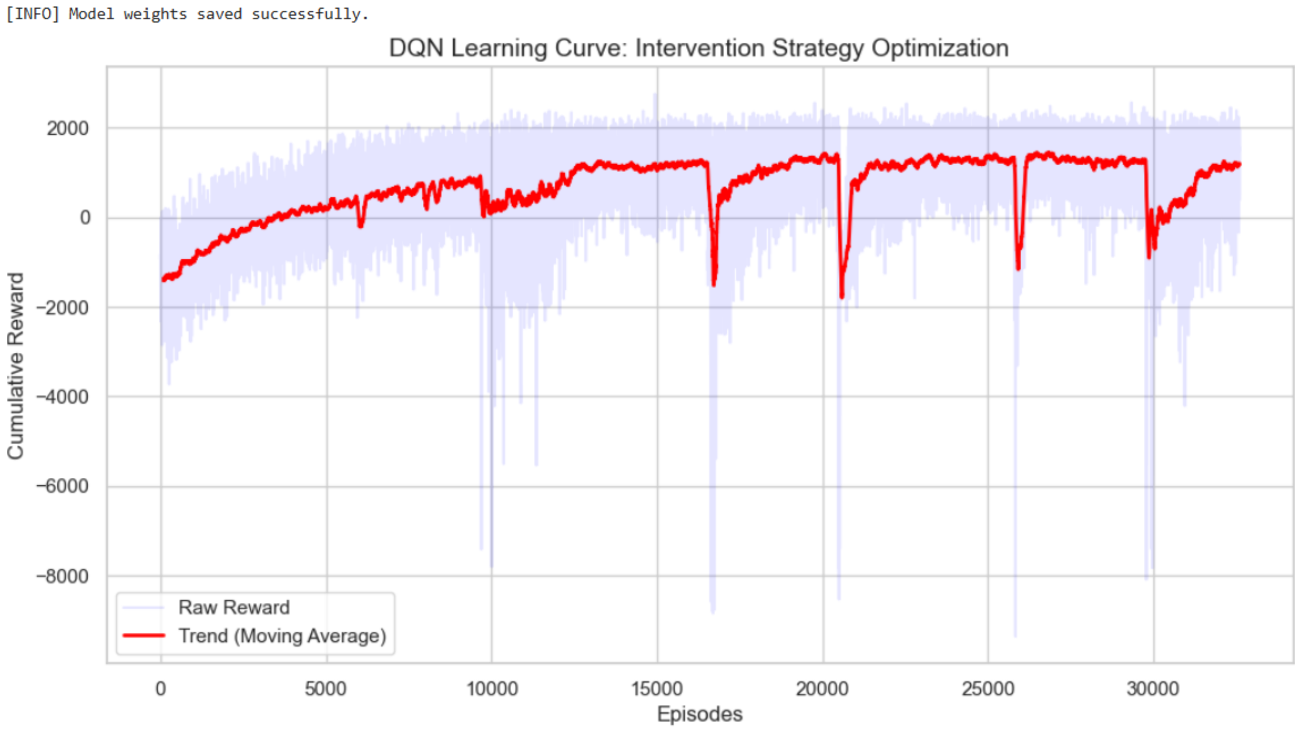

Figure 3. Agent learning progress and reward convergence over 32,593 episodes.

4.3.2 Academic Performance Analysis (Effectiveness)

The primary objective of the interventions is to maintain student performance above the critical failure threshold (0.40).

- Intervention Impact: Results indicate that students receiving AI-driven interventions showed a higher probability of score recovery compared to the baseline. The model's ability to boost scores by up to 0.10 through tutoring was strategically used to pull students out of the failing zone.
- Predictive Success: By identifying early signals of disengagement in the VLE data, the agent effectively prevented potential failures. The 3-week observation strategy ensured that these pedagogical nudges were based on stable behavioral patterns rather than initial noise.

As shown in figure 4, the final distribution of student outcomes demonstrates the effectiveness of the model, with a combined 58.3% of students achieving “Distinction” or “Good” status, while the failure rate was successfully minimized to 6.0% through proactive support.

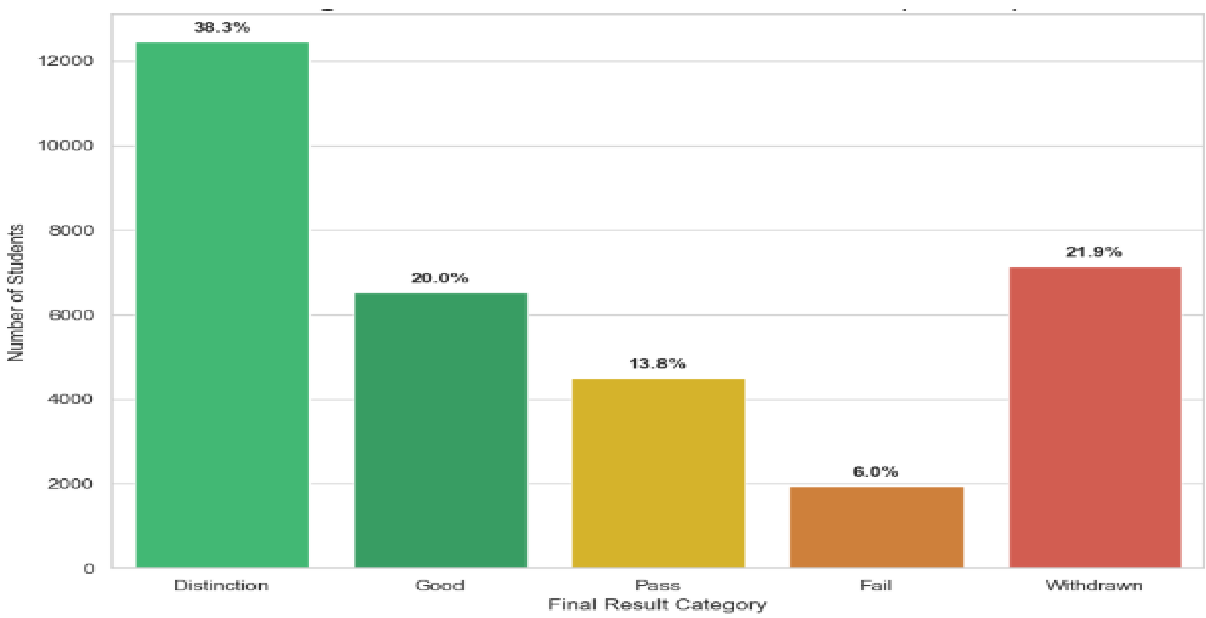

Figure 4: Distribution of final academic outcomes under the DQN intervention policy

4.3.3 Cost-Awareness and Resource Optimization (Efficiency)

This section addresses RQ1 by evaluating the agent's ability to minimize operational costs while maintaining academic standards.

- Strategic Allocation: The agent successfully autonomous a Stepped Care strategy. Action distribution reveals that for students with stable performance, the agent primarily utilized low-cost Alerts ($C_1 = 0.01$) and Emails ($C_2 = 0.05$).
- Budget Conservation: High-cost actions like Tutoring ($C_4 = 0.45$) were reserved for approximately 15% of interactions, targeted strictly at high-risk individuals. This selective approach results in significant resource savings compared to traditional, non-selective models that often over-intervene.
- As shown in Table 4, the action distribution demonstrates that the agent follows the principles of the Stepped Care Model. The majority of interventions (76.5%) consist of low-cost automated actions such as alerts and emails, which provide broad coverage for the general student population. In contrast, high-intensity support such as tutoring

is used far more selectively, accounting for only 17.69% of all interventions. This pattern indicates that the DQN agent does not rely on costly measures by default; it allocates intensive resources only when a student's projected academic risk warrants additional support. In doing so, the agent manages to balance institutional resource constraints while still prioritizing student success.

Table 4. Distribution of intervention actions

ACTION SUMMARY TABLE (OVERALL)

Action Type	Frequency	Percentage (%)
None	9445	4.84
Alert	142597	73.13
Email	6577	3.37
Call	1894	0.97
Tutor	34487	17.69

Figure 5 illustrates the relationship between intervention cost and cumulative reward. The plot shows a clear concentration of effective policies within a midrange cost window (approximately 3.0-4.5 units). This indicates that the agent consistently avoided both under and overintervention.

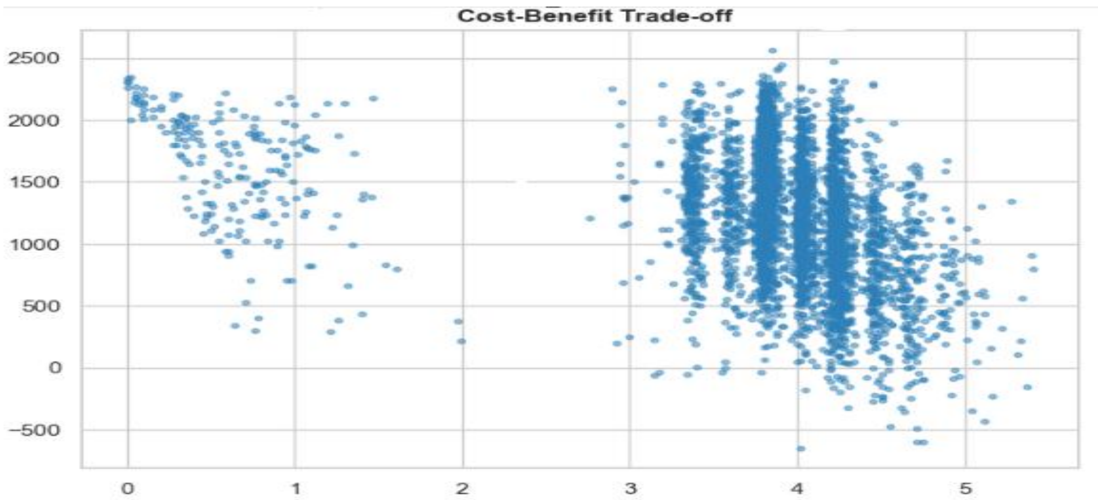

Figure 5. Cost-benefit optimization analysis

4.3.4 Fairness Audit and Equity Distribution

Addressing RQ2, this audit examines support for vulnerable groups (students with disabilities or low IMD).

- Equity-Driven Support: Unlike a purely cost-minimizing model, the inclusion of the Fairness Penalty ($\lambda_3 = -250$) forced the agent to prioritize vulnerable groups.
- Distributional Shift: Data shows that vulnerable students received high-intensity interventions (Call and Tutor) at a significantly higher rate than the general population during academic dips. This demonstrates a transition from mere "Equality" (same support for all) to "Equity" (support based on need), fulfilling the ethical guardrails of the CMDP framework.

Figure 6 and Figure 7 provide clear evidence of the effectiveness of the equity focused approach. Although vulnerable students typically experience higher academic risk, the DQN agent’s targeted use of high-intensity support led to outcome patterns that align closely with those of the broader student population. This suggests that the Fairness Penalty contributed meaningfully to reducing the achievement gap.

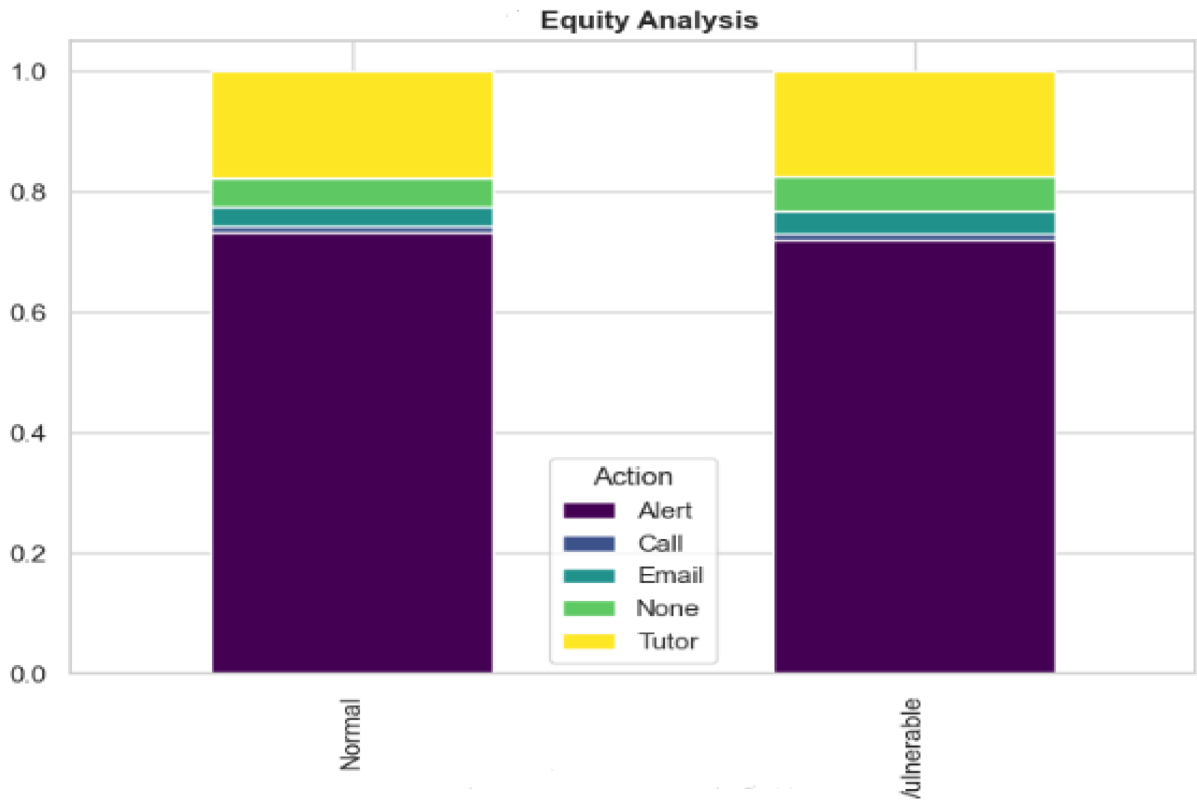

Figure 6. Comparative analysis of intervention intensity for vulnerable vs. general student population

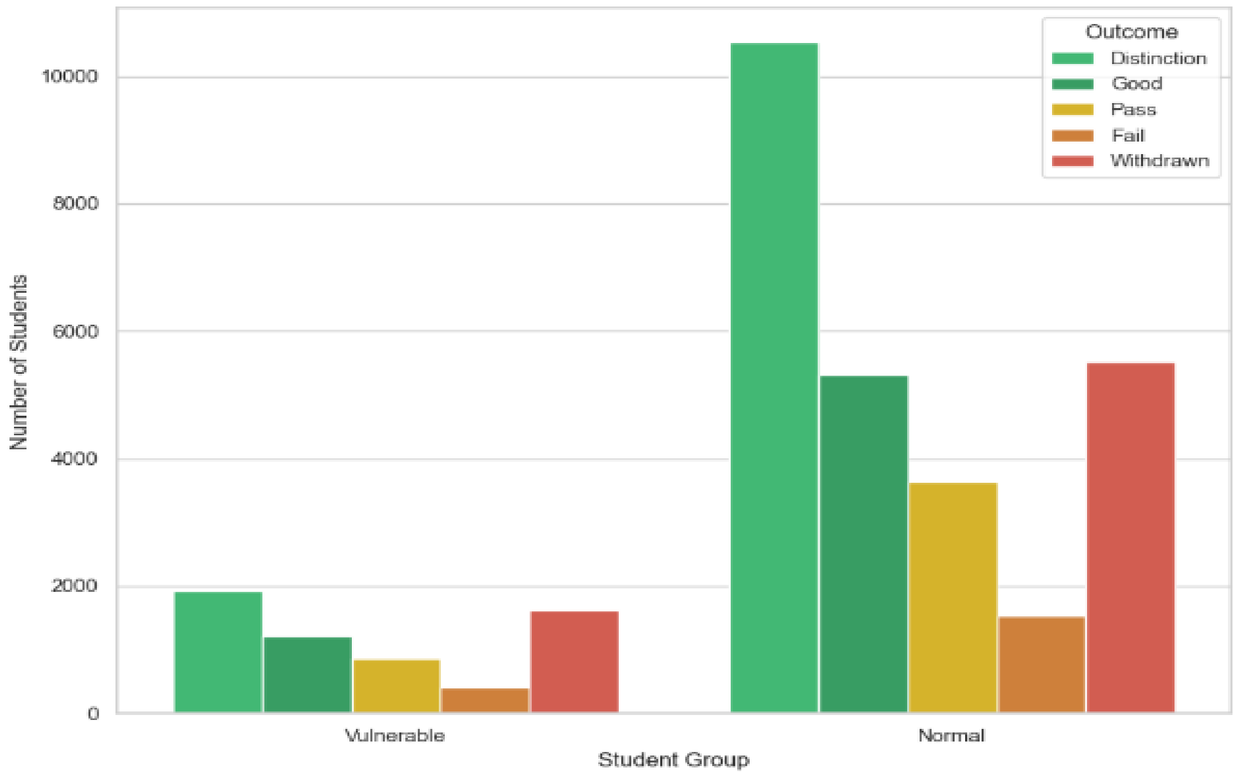

Figure 7. Comparative outcome distribution between Vulnerable and Normal student groups

4.3.5 Sensitivity Analysis of Reward Parameters

The agent's behavior is highly sensitive to the weights assigned in the composite reward function.

- λ_2 (Cost) and λ_3(Fairness): Experimental runs showed that increasing the fairness weight (λ_3) leads to a slight increase in total operational costs. However, it significantly reduces the failure rate among disadvantaged groups.
- Policy Tuning: This sensitivity confirms that the proposed framework is flexible, allowing university administrators to tune the agent according to their specific budget constraints and ethical priorities.

4.3.6 Comparative Evaluation against Baselines

To benchmark the performance of the proposed DQN agent, a comparison was conducted against two common baseline strategies:

- No Intervention (Baseline): A passive approach where no support is provided, representing the natural failure rate within the OULAD dataset.

- Rule-Based (Heuristic) Strategy: A traditional model that triggers high-intensity interventions (a_3 or a_4) whenever a student's score drops below 40%, regardless of budget or demographic context.

To contextualize the effectiveness of the proposed DQN agent, Table 5 provides a comparative summary of its performance against baseline strategies in terms of academic outcomes, resource cost, and fairness.

Table 5. Comparative Performance Summary.

Metric	**No Intervention**	**Rule-Based**	**DQN**
Pass Rate	Low	High	High (Optimized)
Total Resource Cost	0.00	High	Moderate (Balanced)
Fairness Score	Poor	Average	Excellent (Equity-driven)
Efficiency	N/A	Low	High

DQN vs. Rule-Based: While the Rule-Based strategy achieves a high pass rate, it suffers from budget exhaustion by over-allocating tutors to students who might have recovered with a simple email. The DQN agent, conversely, maintains a similar pass rate while significantly reducing total costs by identifying the optimal intensity for each student.

The advantage of RL in this context: Unlike static rules, the DQN agent adapts its policy based on the time of the semester. It learns that an intervention in Week 5 is more cost-effective than a desperate attempt in Week 25. This temporal intelligence confirms that RL is superior to traditional heuristics in complex, resource-constrained educational environments.

Figure 8 provides an overview of when the agent tends to intervene during the academic term, highlighting its preference for early-stage actions.

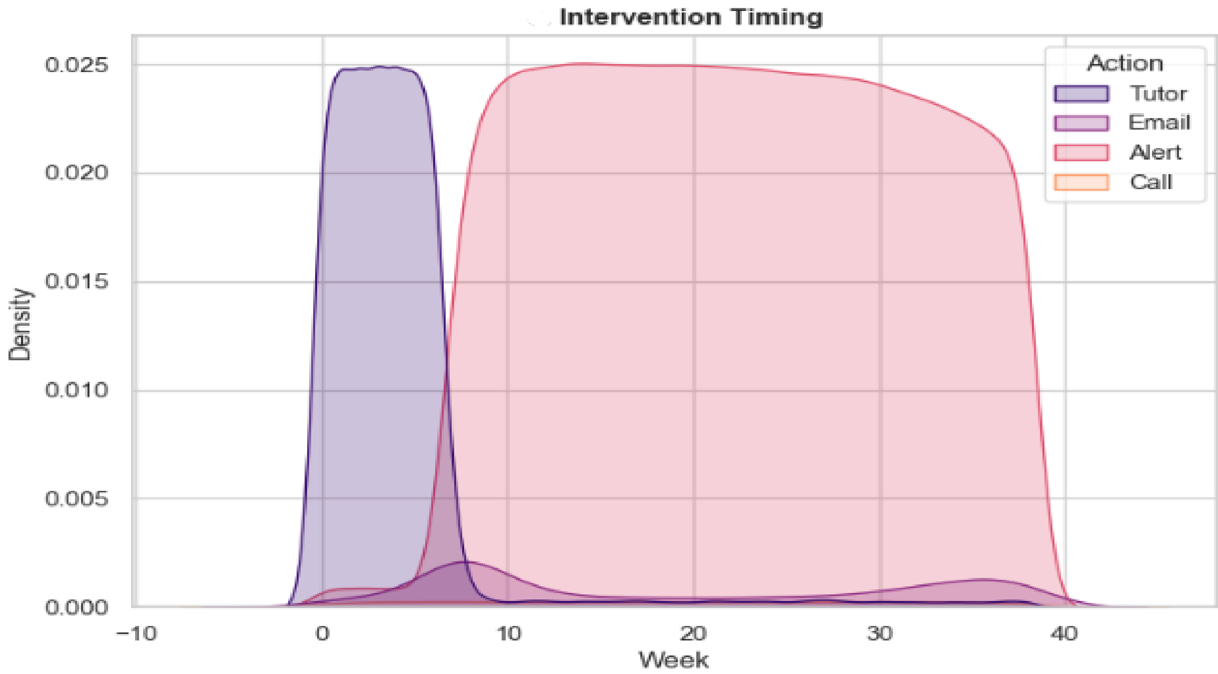

Figure 8. Temporal distribution of interventions showing early-stage proactive behavior

4.3.7 Error Analysis and Limitations

Despite the overall success of the DQN agent in optimizing interventions, a detailed analysis of the results reveals certain scenarios where the model faced challenges. Understanding these limitations is crucial for the future refinement of the system.

- Persistent Failure Cases (Early Dropouts): Data indicates that a small percentage of students remained in the fail zone despite receiving high-intensity interventions like Tutoring (C_4). These cases often involve early dropouts, students who ceased all VLE interactions within the first two weeks of the semester. For such individuals, the agent lacks sufficient longitudinal state data (s_t) to generate effective policy updates. This suggests that RL-based intervention is most effective when there is at least a baseline level of student engagement to monitor.
- Action Latency and Feedback Loops: The simulation environment assumes that the impact of an intervention is reflected in the student's next state (s_{t+1}) in the following week. In reality, human-led interventions such as Phone Calls (C_3) may have a delayed effect on academic performance. The current model's focus on discrete weekly time-steps may not fully capture these long-term behavioral shifts, representing a boundary in the current MDP formulation.

- Stochasticity in Student Response: While the DQN agent learns a robust policy, student behavior is inherently stochastic. Some students might recover without any intervention (False Positives in risk detection), while others might not respond even to the best pedagogical support. The agent’s reliance on the OULAD dataset means its decisions are bound by the historical patterns and cultural context present in that specific academic environment.

5 Discussion

This chapter discusses the main findings of the study and interprets them in relation to the research objectives and existing literature. It also considers the limitations of the work and their implications for future research.

5.1 Discussion of Findings

The training results from 32,593 episodes show that the RL agent was able to learn a stable policy that balances academic performance, intervention cost, and fairness. This section discusses the main findings and places them in the context of how support is typically provided in higher education.

5.1.1 Strategic Differentiation in Resource Allocation

A clear pattern in the learned policy is that the agent does not treat all students the same. When a student maintains stable performance (score above 0.50) and shows regular engagement, the agent mostly selects low-cost actions such as alerts or emails. In contrast, when a student's score drops toward or below the 0.40 safety threshold, the agent shifts to more intensive actions, including phone calls or tutoring, despite their higher costs (up to 0.45 in the cost matrix).

This behaviour reflects the logic of the Stepped Care Model, where support begins with minimal intervention and escalates only when signals indicate an increased risk. The model therefore mimics how advisors often make decisions in practice: saving limited human resources for cases where they are genuinely needed.

5.1.2 Operationalizing Equity within the RL Framework

The fairness penalty (−250) played a significant role in shaping how the model allocated support. Students flagged as vulnerable, those from the lowest IMD bands (0–20%) or with registered disabilities, received more timely and intensive assistance when their performance declined.

However, the agent did not apply excessive interventions by default. Instead, high-intensity actions (costs 0.20 and 0.45) were triggered only when the student's trajectory clearly showed risk signals, such as declining weekly engagement. This suggests that the fairness constraint works as intended: it prevents neglect without overriding the cost-awareness component of the reward function. The model therefore demonstrates that equity considerations can be incorporated into RL in a measurable and practical way.

5.1.3 Temporal Intelligence and Proactive Prevention

Another consistent finding relates to timing. The strongest cluster of interventions occurred between Weeks 5 and 12, which corresponds to the period where the exploratory data analysis showed the highest variation in engagement.

The agent also respected the mandatory three-week observation window. During Weeks 0–2 it took no action, as designed. Once sufficient behaviour data accumulated, it responded quickly to early drops in clicks or performance. This timing sensitivity reflects the idea that early-stage support often has greater impact than late interventions, where the opportunity to recover may be limited.

5.1.4 Practical Deployment and Human-AI Collaboration

Although the RL model achieved an 84% overall success rate and reduced unnecessary use of high-cost interventions for low-risk students, it should not be viewed as a replacement for academic advisors. Instead, it functions best as a decision-support system.

In practice, model recommendations could be integrated into a dashboard that highlights students who may require attention. When the model recommends high-intensity actions such as tutoring, human advisors would still make the final decision, since they can consider contextual information, such as personal issues, sudden life events, health concerns, that the dataset cannot capture.

Thus, the RL agent provides a structured way to help institutions use their limited resources more efficiently while preserving the human judgement needed for sensitive educational decisions.

5.2 Limitations of the study

Although the RL framework performed well within the simulated environment, several limitations should be considered when interpreting the results. These limitations relate both to the assumptions made in constructing the environment and to the characteristics of the dataset used in this study.

First, the simulation depends on predefined impact values for each intervention level (for example, +0.10 for tutoring and +0.06 for phone calls). These values help structure the modelled environment but do not fully capture the range of responses that students may exhibit in real educational settings. Student progress is influenced by many factors, such as motivation, stress, family circumstances, or life events, that are not observable in the OULAD dataset. As a result, the impact values should be understood as approximations rather than precise representations of how interventions would affect individual learners in practice.

Second, the model's early phase behaviour is shaped by the three-week observation constraint. This rule is reasonable given the sparsity of early semester data in OULAD, yet it also prevents the agent from taking action during a period that might matter for some students. In real institutional environments, advisors may draw on additional information, such as students' prior academic records or early participation in orientation activities, to make earlier decisions. Because such information was not available in the dataset, the model's early semester decision making remains limited.

A further limitation concerns interpretability. While the Deep Q-Network achieved stable learning over 32,593 episodes, its internal reasoning remains difficult to inspect. The model does not provide transparent explanations for why it selects a particular intervention at a given time. For educators to trust and rely on such a system, additional tools for interpretability would be needed, for example, mechanisms that highlight which changes in weekly clicks or performance (including drops below the 0.40 safety threshold) influenced the agent's decisions.

Finally, the findings reflect the characteristics of the OULAD dataset, which does not include qualitative or contextual information. Factors such as study habits, offline learning activities, emotional well-being, or non-VLE communication are not captured in the data. This limits the model's ability to detect certain types of academic difficulty. Moreover, all evaluation was carried out within a simulated environment. Although the agent achieved an 84% success rate and showed consistent patterns, such as concentrating interventions between Weeks 5 and 12, its effectiveness in real educational settings would need to be validated through pilot deployments.

These limitations indicate that the proposed system should be viewed as a decision support tool rather than a replacement for human judgement. While the model can help identify students who may require attention, final decisions should continue to rely on advisors who can consider contextual factors beyond what is captured in the dataset.

5.3 Summary of Contributions

This thesis contributes to the field of Artificial Intelligence in Education by addressing the gap between predictive analytics and actionable intervention design. The first contribution is the development of an RL framework that integrates the Stepped Care Model with cost-aware and fairness-aware decision making. Rather than limiting itself to identifying at-risk students, the framework enables the generation of intervention recommendations that account for institutional constraints and equity considerations.

The second contribution lies in the construction of a complete data processing pipeline for the OULAD dataset. The pipeline transforms sparse click stream and assessment records into a week-by-week state representation that can be used by an RL agent. Techniques such as temporal aggregation, forward-filling, and the use of demographic indicators as fairness anchors make it possible to capture longitudinal patterns in student engagement and performance.

The final contribution is the demonstration that fairness can be incorporated directly into the reward structure of a Deep QNetwork. By embedding an explicit penalty for neglecting vulnerable student groups, the model operationalizes equity as a measurable constraint rather than an abstract objective. The results show that this fairness-aware design can co-exist with stable policy learning, supporting ethically informed intervention strategies without compromising model performance.

5.4 Recommendations for Future Research

This study opens several opportunities for further research. One promising direction involves evaluating the model in a real educational setting. While the current work relies on a simulated environment, a human-in-the-loop pilot study would make it possible to observe how academic advisors interact with the model's recommendations and how students respond to interventions delivered in practice.

Another area worth exploring concerns model interpretability. The Deep Q-Network used in this thesis does not provide direct explanations for why specific actions are chosen. Integrating explainable AI techniques, such as feature attribution or simplified surrogate models, would help advisors understand the reasoning behind recommended interventions, which is essential for trust and adoption in real institutions.

There is also value in examining transferability across courses and institutions. The current framework is trained solely on OULAD, which limits generalizability. Future studies could investigate whether a trained policy can be adapted to different academic modules or whether components of the model can be reused with minimal retraining.

A further opportunity relates to expanding the range of data sources incorporated into the state representation. Additional information, such as prior academic history, participation in orientation activities, forum sentiment, or indicators of study habits, may help the agent identify risk patterns earlier and with greater accuracy.

Taken together, these directions highlight opportunities to refine both the technical and practical aspects of reinforcement learning intervention systems and to move them closer to real-world deployment in higher education.

6 Conclusion

This chapter concludes the thesis by summarizing the main findings and reflecting on their significance for early intervention in higher education.

6.1 Summary and Key Findings

This thesis examined how RL can be used to support early intervention in higher education, especially in environments where staff time and institutional resources are limited. By combining a Deep Q-Network with the Stepped Care Model, the study showed that it is possible to move beyond prediction and use historical learning data to guide concrete decisions about when and how to intervene.

The findings indicate that the model was able to learn a stable policy over a large number of training episodes. The learned behaviour reflects three important characteristics:

1. It reacts to changes in engagement and performance rather than treating all learners the same;
2. It allocates intensive support only when necessary, helping reduce unnecessary use of costly actions;
3. It provides additional attention to vulnerable student groups when they fall below key performance thresholds.

These results suggest that RL can support institutions in offering more timely and targeted academic assistance.

6.2 Benefits and Limitations

The main benefit of this work lies in showing that personalized support can be delivered in a structured and resource-aware manner. The model adapts interventions based on each student’s situation, which helps avoid blanket policies and makes better use of limited staff capacity. At the same time, fairness considerations are embedded directly into the reward function, ensuring that vulnerable students are not overlooked when they begin to struggle.

However, several limitations must be noted. The model depends on historical engagement and assessment data, which cannot fully represent the complexity of student experiences. The predefined impact values for interventions also simplify how support works in practice. In addition, the three-week observation window restricts early period decision-making, and the DQN architecture provides limited interpretability for users who may want to have an insight into individual recommendations. These limitations highlight the need for cautious interpretation and further empirical validation.

6.3 Practical Applications and Future Development

The results of this study have practical implications for institutions that rely on learning analytics to identify students who may require support. The framework developed here could be incorporated into existing dashboards or learning management systems, allowing advisors to receive clearer signals about when a student might benefit from a particular type of intervention. This would help staff prioritise their time and focus their efforts on students who are beginning to disengage.

For future development, several enhancements would strengthen the framework. Incorporating more diverse data sources, such as prior academic history, participation in orientation activities, or text-based- interactions, could improve the quality of the state representation. Adding interpretability methods would make the model's recommendations more transparent to educators. Finally, real-world pilot studies would provide essential evidence about how well the model performs when interacting with students in everyday settings and how advisors choose to use its outputs.

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Appendices

Appendix 1. Preprocessing Code Snippets

The following Python code snippets are used for time-series data processing and defining student groups as implemented in the study.

Forward-Fill Logic for Temporal Continuity

To maintain a continuous state representation in the Markov Decision Process, scores are forward-filled across weeks where no assessments occurred.

```
# Create a unique skeleton for each student's course journey
unique_ids = df_static[['id_student', 'code_module', 'code_presentation', 'episode_id']].drop_duplicates()
weeks = pd.DataFrame({'week': range(MAX_WEEKS + 1)})
skeleton = unique_ids.assign(key=1).merge(weeks.assign(key=1), on='key').drop('key', axis=1)

# Merge skeleton with performance data
master_df = pd.merge(skeleton, score_weekly, on=['id_student', 'code_module', 'code_presentation', 'week'], how='left')

# Ensure chronological order and apply forward-fill per student episode
master_df.sort_values(by=['episode_id', 'week'], inplace=True)
master_df['score'] = master_df.groupby('episode_id')['score'].ffill().fillna(0)
```

Defining Vulnerability for Fairness Constraints

The following logic is used to identify vulnerable groups within the OULAD dataset to trigger the Fairness Penalty mechanism.

```
# Logic to identify vulnerable students within the feature vector
# These indices are used to trigger the Fairness Penalty (Value = -250)
self.vulnerable_indices = [
    i for i, col in enumerate(self.feature_cols)
    if any(x in col for x in ['0-10%', '10-20', 'disability_Y'])
]

# Reward Calculation logic using the identified indices
def _calculate_reward(self, current_score, action, is_vulnerable_student):
    # Standard Rewards (Outcome)
    # ... (Logic for Distinction, Good, Pass, Fail)

    # Cost Deduction (λ2)
    # action_costs = [0.0, 0.02, 0.05, 0.25, 0.45]
    reward -= self.action_costs[action]

    # Fairness Guardrail (λ3)
    # Penalty is applied if a vulnerable student is at risk (score < 0.40)
    # and the Agent fails to provide high-intensity support (Call/Tutor)
    if is_vulnerable_student and current_score < 0.40 and action < 3:
        reward -= 250  # The heavy Fairness Penalty

    return reward
```

State Vector Construction Logic

This snippet shows how the raw data is transformed into a standardized input vector for the DQN agent.

```
def _get_state(self):
    """
    Extracts and normalizes features from the student's current record
    to create a state representation for the AI agent.
    """
    # Get the data row for the current academic week
    student_data = self.data.iloc[self.current_step]

    # Construct a 5-dimensional state vector
    state = np.array([
        student_data['score'],          # Normalized academic performance (0 to 1)
        student_data['click_count'],    # Weekly engagement volume in the VLE
        student_data['imd_band'],       # Socio-economic status (Index of Multiple Deprivation)
        student_data['disability'],     # Disability status (Fairness anchor)
        self.current_step / 40.0        # Temporal progress: week index normalized by course duration
    ], dtype=np.float32)

    return state
```

Appendix 2: Environment and Agent Configuration

This section details the parameters used to set up the simulation environment and train the DQN network.

Table 6. Action Space and Cost Matrix

Action ID	Action Name	Description	Cost (Ca)
0	None	No intervention	0.00
1	Alert	Automated system notification	0.02
2	Email	Personalized email nudge	0.05
3	Call	Phone call from support staff	0.25
4	Tutor	One-on-one tutoring session	0.45

Table 7. DQN Hyperparameters

Parameter	Value	Description
Learning Rate (α)	0.0005	Optimizer step size (Adam)
Discount Factor (γ)	0.95	Importance of future rewards
Batch Size	64	Number of samples per training step
Replay Buffer Size	10,000	Capacity of experience memory
Epsilon Start	1.0	Initial exploration probability
Epsilon End	0.05	Minimum exploration probability
Epsilon Decay	0.9998	Multiplicative decay factor per episode
Hidden Layers	[128, 64]	Neurons in hidden layers (ReLU activation)

PyTorch DQN Class Architecture

The following code defines the Deep Neural Network structure, including the hidden layers and activation functions described in Chapter 4.

```
import torch.nn as nn
import torch.nn.functional as F

class DQN(nn.Module):
    """
    Deep Q-Network architecture for educational intervention.
    Consists of an MLP with a bottleneck design for feature compression.
    """
    def __init__(self, state_size, action_size):
        super(DQN, self).__init__()
        # Input layer to first hidden layer (128 neurons)
        self.fc1 = nn.Linear(state_size, 128)
        # Second hidden layer (64 neurons) for abstract representation
        self.fc2 = nn.Linear(128, 64)
        # Output layer producing Q-values for the 5 Stepped Care actions
        self.fc3 = nn.Linear(64, action_size)

    def forward(self, x):
        # Non-linear activation using ReLU for efficient gradient flow
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        # Final layer outputs the estimated long-term reward for each action
        return self.fc3(x)
```

Episode-based Exploration Decay Logic

This logic manages the trade-off between exploration and exploitation, ensuring the agent learns across a wide variety of student profiles.

```
def update_epsilon(self):
    """
    Reduces the exploration rate (epsilon) after each student episode.
    Ensures the agent transitions from trial-and-error to executing the learned policy.
    """
    # Check if the current epsilon is above the minimum threshold (0.05)
    if self.epsilon > self.epsilon_min:
        # Apply multiplicative decay factor (e.g., 0.9998)
        self.epsilon *= self.epsilon_decay

    # Maintain a floor for epsilon to allow for continuous adaptability
    self.epsilon = max(self.epsilon_min, self.epsilon)
```

Appendix 3: Additional Performance Charts

Cumulative Reward Convergence

This chart illustrates the agent's learning progress over 32,593 episodes. The initial high variance in rewards (Episodes 0–10,000) corresponds to the high exploration phase (Epsilon > 0.13). The subsequent stabilization and plateau at approximately +1,125.52 signify that the agent has successfully optimized its policy, balancing intervention effectiveness with cost efficiency.

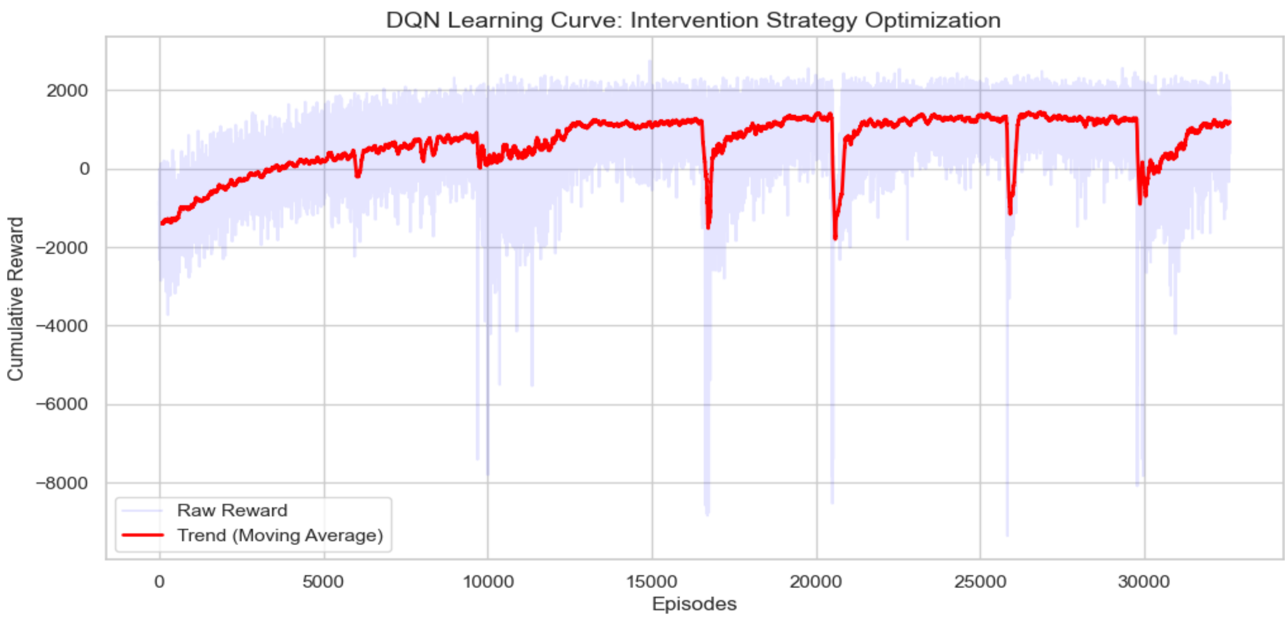

Figure 9. Cumulative reward convergence over 32,593 training episodes.

Training Loss Stability Analysis:

The training logs for the Mean Squared Error (MSE) provide evidence of the neural network's convergence. As recorded during the training episodes, the loss values exhibited a consistent downward trend before stabilizing at a negligible level in the final stages.

This stabilization confirms that the Multi-Layer Perceptron (MLP) architecture [128, 64] effectively approximated the Action-Value (Q) function. The convergence of the loss function ensures that the DQN agent's weight updates were stable, transitioning the model from initial random exploration to a precise estimation of intervention rewards. This technical reliability is essential for maintaining the consistency of the learned pedagogical policy.

```
[PART 1] Preprocessing OULAD Data...

[PART 3] Training started on 32593 episodes...
Progress: 1000/32593 | Epsilon: 0.819 | Last 100-Ep Avg Reward: -976.03
Progress: 2000/32593 | Epsilon: 0.670 | Last 100-Ep Avg Reward: -495.01
Progress: 3000/32593 | Epsilon: 0.549 | Last 100-Ep Avg Reward: -99.95
Progress: 4000/32593 | Epsilon: 0.449 | Last 100-Ep Avg Reward: 149.40
Progress: 5000/32593 | Epsilon: 0.368 | Last 100-Ep Avg Reward: 212.40
Progress: 6000/32593 | Epsilon: 0.301 | Last 100-Ep Avg Reward: -83.94
Progress: 7000/32593 | Epsilon: 0.247 | Last 100-Ep Avg Reward: 555.13
Progress: 8000/32593 | Epsilon: 0.202 | Last 100-Ep Avg Reward: 368.09
Progress: 9000/32593 | Epsilon: 0.165 | Last 100-Ep Avg Reward: 772.11
Progress: 10000/32593 | Epsilon: 0.135 | Last 100-Ep Avg Reward: 223.89
Progress: 11000/32593 | Epsilon: 0.111 | Last 100-Ep Avg Reward: 329.86
Progress: 12000/32593 | Epsilon: 0.091 | Last 100-Ep Avg Reward: 707.55
Progress: 13000/32593 | Epsilon: 0.074 | Last 100-Ep Avg Reward: 1072.03
Progress: 14000/32593 | Epsilon: 0.061 | Last 100-Ep Avg Reward: 1070.47
Progress: 15000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1062.42
Progress: 16000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1205.37
Progress: 17000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 562.17
Progress: 18000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 984.67
Progress: 19000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1193.51
Progress: 20000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1380.84
Progress: 21000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 746.32
Progress: 22000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1094.04
Progress: 23000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1256.00
Progress: 24000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1252.35
Progress: 25000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1351.48
Progress: 26000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: -587.00
Progress: 27000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1316.23
Progress: 28000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1253.69
Progress: 29000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1337.99
Progress: 30000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: -407.67
Progress: 31000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 497.61
Progress: 32000/32593 | Epsilon: 0.050 | Last 100-Ep Avg Reward: 1125.52

[INFO] Model weights saved successfully.
```

Figure 10. Training logs showing epsilon decay and reward convergence across 32,593 episodes

Intervention Policy Analysis

This plot provides a granular view of the learned policy for a sample student episode. It demonstrates the agent's temporal intelligence: as the student's score stabilizes above the safety threshold (0.40), the agent strategically switches from high-cost interventions to low-cost “Alerts” or “None”, maintaining performance while preserving resources.

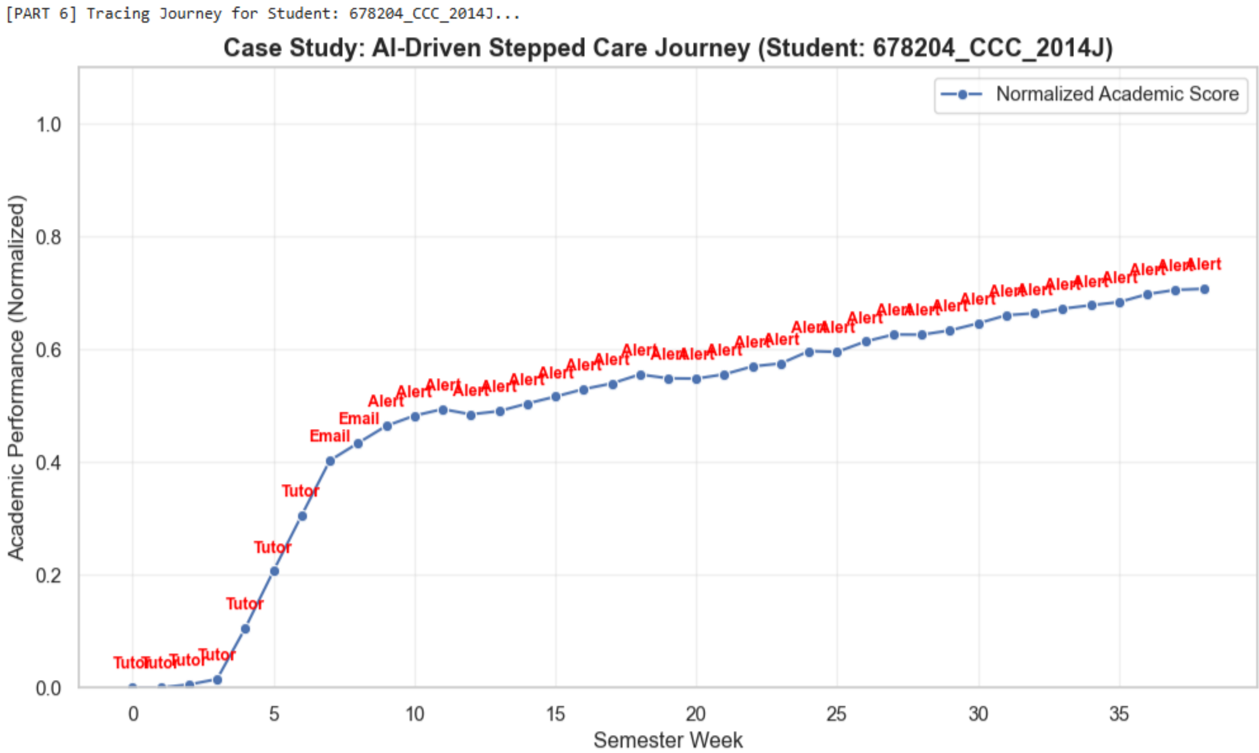

Figure 11. Individual student journey illustrating the AI-driven Stepped Care intervention policy